

# **Does Personalization Change Consumer Responses to AI-Authored Marketing Messages?**

AI Authorship, Authenticity, and Consumer Response in Marketing Communications

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## **Abstract**

This thesis examines whether personalizing email content changes consumer responses to AI-attributed emotional marketing communication. Prior research has found that disclosure of AI authorship in emotional marketing communications can reduce positive word of mouth and customer loyalty. Kirk and Givi (2025) refer to this pattern as the AI-authorship effect.). This study therefore tests whether personalizing the email content itself, rather than rewording the disclosure, changes consumer responses to AI-authored communication. A  $2 \times 2$  between-subjects experiment ( $N = 79$ ) assigned participants to a human- or AI-attributed loyalty email with generic or personalized content from a fictional fashion brand. Observed means differed across conditions: Human–Generic showed higher PWOM and loyalty than AI–Generic, while AI–Generic showed slightly higher authenticity. AI–Personalized had the highest observed loyalty, whereas Human–Personalized showed higher PWOM and authenticity than AI–Personalized. However, the direct comparisons and authorship  $\times$  personalization interactions did not provide statistically reliable evidence for these differences. Interpretation is constrained by weak recognition of the manipulations and the smaller-than-planned sample. The findings highlight the importance of communication context and manipulation salience when evaluating responses to AI-attributed emotional marketing communication.

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## 1. Introduction

Artificial intelligence is increasingly used in marketing operations, with organizations applying generative AI tools to produce emails, social media posts, and other forms of customer-facing communication at scale (Chandra et al., 2022). For large brands managing customer relationships simultaneously across millions of consumers, this development raises questions that extend beyond operational efficiency.

One of the most consequential of these questions concerns consumer perception: how do consumers respond when they know a message was written by an artificial intelligence? This matters because marketing communications do not only convey information; they can also signal relational investment. An email expressing appreciation or loyalty can lead consumers to believe that the brand is making an effort and showing care. When an AI system generates that message, consumers may infer that the emotional expression is less directly based on human emotions, which can affect authenticity judgments and behavioral intentions (Morhart et al., 2015; Kirk & Givi, 2025).

Recent empirical work has begun to examine this question. Kirk and Givi (2025) report across seven experiments that AI authorship of emotional marketing communications was associated with lower positive word of mouth and customer loyalty. They identify perceived authenticity as an important explanatory construct in this pattern, which they refer to as the AI-authorship effect. Schilke and Reimann (2025) extend this inquiry by testing whether varying the framing of the disclosure can reduce the cost. Across thirteen rhetorical framings, their study found that changing how AI involvement was described did not eliminate the trust reduction relative to a no-disclosure baseline. These findings suggest that the content and context of communication may matter in addition to the wording of the disclosure.

These findings have practical implications for organizations using generative AI in customer communications. If rhetorical reframing does not reduce the response cost of AI disclosure, organizations may need to consider whether other dimensions of the communication itself can make a difference. One candidate dimension is personalization, which may signal individual knowledge of the recipient rather than merely changing how AI involvement is disclosed. The present study therefore asks whether email content that contains mild and generalized behavioral references can alter consumer responses to AI-authored communication.

The literature reviewed here does not appear to have directly tested this question. The existing personalization literature suggests that content adapted to individual behavioral histories can increase engagement and signal consumer orientation (Chandra et al., 2022) but also identifies conditions under which personalization may backfire by triggering privacy concerns and perceptions of intrusiveness (Aguirre et al., 2015). Whether these dynamics interact with AI authorship attribution in ways that alter authenticity perceptions remains an open empirical question. The present study aims to provide initial evidence addressing this question.

This study employs a between-subjects experiment. Participants read an emotional loyalty email from a fictional fashion brand under one of four conditions and reported their perceptions of the communication's authenticity and their behavioral intentions toward the brand. The study tests whether the Human–Generic versus AI–Generic comparison is detectable in a new sample and context, and whether content personalization moderates the relationship between authorship and consumer responses.

The study examines whether the AI-authorship effect is detectable in a new sample and communication context, while also documenting how manipulation salience and measurement quality may shape the observed consumer responses.

The thesis is organized as follows. Section 2 reviews the relevant literature. Section 3 develops the theoretical framework and states the hypotheses. Section 4 describes the experiment. Section 5 presents the results. Section 6 provides general discussion, including theoretical contributions, practical implications, limitations, and future research directions. Section 7 concludes the thesis.

Generative AI tools were used during the development of this thesis for language editing, proofreading, feedback on structure and clarity, and assistance with selected analytical and coding tasks. AI assistance was used to draft and troubleshoot R code for data cleaning, measure scoring, descriptive statistics, reliability checks, hypothesis tests, supplementary analyses, sensitivity calculations, and figure generation. The author remained responsible for all theoretical, methodological, analytical, and interpretative decisions and verified the final content, references, analyses, and conclusions. The anonymized data, data dictionary, survey materials, analysis code, statistical outputs, documentation of substantial GenAI interactions, and additional documentation associated with this study are available through the project's Open Science Framework (OSF) repository: <https://osf.io/tbquh/overview>

## **2. Literature Review**

### **2.1 Brand Authenticity in Marketing Communications**

Authenticity has emerged as one of the most important constructs in consumer psychology and brand management research. It has been defined as the perception that an entity is genuine, true to its own values, and free from purely instrumental or commercial motivation (Morhart et al., 2015; Napoli et al., 2014). In marketing communications, authenticity judgments are particularly relevant because consumers are systematically exposed to persuasive messages designed to elicit favorable responses, creating a natural tension between promotional intent and perceived sincerity.

Morhart, Malar, Guèvremont, Girardin, and Grohmann (2015) develop an integrative theoretical framework for brand authenticity comprising four dimensions: continuity, the degree to which a brand is consistent over time and true to its origins; credibility, whether the brand delivers on its promises; integrity, the extent to which the brand acts from genuine values rather than purely financial self-interest; and symbolism, the capacity of the brand to help consumers express authentic aspects of their own identity. Of these, the integrity dimension is most directly implicated in the evaluation of marketing communications. When a consumer receives an emotional message from a brand, such as a loyalty email expressing gratitude, the central authenticity question is whether this specific act of communication reflects genuine values and relational investment rather than a calculated persuasion attempt.

Perceived authenticity in the present study is operationalized using the four-item bipolar scale developed by Napoli, Dickinson, Beverland, and Farrelly (2014) and subsequently employed by Kirk and Givi (2025), capturing integrity-relevant judgments including the distinction between inauthentic and authentic, and between compromising and not compromising one's values. This operationalization aligns with the integrity dimension of

Morhart et al.'s framework, which is the dimension most directly implicated when consumers question whether brand communication is genuine.

## **2.2 The AI-Authorship Effect**

The growing use of generative AI in commercial contexts introduces a new type of authenticity challenge. Earlier research on brand inauthenticity focused on inconsistency or deceptive advertising. AI-generated communications raise a different concern: can a message authored by a system that consumers do not perceive as having human emotional experience be seen as genuinely expressing the emotions it conveys?

Kirk and Givi (2025) provide the most direct empirical answer. Across seven experiments, they find that AI authorship of emotional marketing communications consistently lowers positive word of mouth and customer loyalty. Perceived authenticity is identified as an important explanatory construct: consumers perceive AI-authored emotional messages as less genuine, and this lower perceived authenticity is associated with less favorable behavioral intentions. The authors call this the AI-authorship effect.

The theoretical argument rests on a distinction between factual and emotional communications. Factual content is less dependent on whether consumers believe that the sender personally experiences the information being communicated. Emotional communications, by contrast, may depend more strongly on the inference that the sender genuinely experiences the emotion being expressed. AI authorship may therefore create a stronger authenticity concern in emotional than in factual communication contexts.

Kirk and Givi (2025) report large effect sizes: in Study 5, perceived authenticity was rated 5.38 in the human condition and 3.99 in the AI condition ( $\eta^2 = 0.18$ ), and the pattern replicated across different emotional valences including gratitude, happiness, sympathy, and pride, as well as across American and British samples.

The effect also generalized across communication contexts. In Study 1, Kirk and Givi tested the effect using a personal salesperson email written to an individual customer after a purchase and found the same pattern of reduced authenticity and lower behavioral intentions under AI authorship. This variation in email format, from a personal post-purchase email in Study 1 to a collective loyalty communication in Study 5, shows that the AI-authorship effect is not limited to one type of emotional communication.

Further findings relevant to the present study come from Study 2, in which the AI-authorship effect was absent for the behavioral outcomes when the message conveyed factual rather than emotional content. The study did not examine perceived authenticity as part of this factual versus emotional comparison. This is consistent with the argument that the AI-authorship effect is more likely to arise for emotional than factual communication, which grounds the choice of an emotional loyalty email as the stimulus in the present study.

### **2.3 AI Disclosure and Consumer Trust**

The AI-authorship effect documented by Kirk and Givi operates under conditions of explicit disclosure. This raises a practical question for organizations: does the way AI involvement is disclosed affect the magnitude of the authenticity penalty? If certain disclosure framings can mitigate the negative response, companies might preserve the efficiency benefits of AI-generated communication while limiting reputational costs.

Schilke and Reimann (2025) provide the most systematic empirical examination of this question, and their findings are largely discouraging. Across thirteen experiments, Schilke and Reimann (2025) tested whether different rhetorical framings of AI disclosure could reduce the trust penalty relative to no disclosure. The framings emphasized strategies such as human oversight, quality assurance, transparency, and the possibility of AI error. None of the framings eliminated the reduction in trust relative to the no-disclosure baseline. This suggests that changing the wording of the disclosure may be

insufficient when consumers are concerned about the role of AI in communication. Their findings indicate that trust may decline across different disclosure formats, although the specific psychological reason for this decline remains open to further investigation.

Consistent patterns appear in the advertising literature. Baek, Kim, and Kim (2026) find that AI disclosure in prosocial advertising reduces perceived ad credibility, which in turn lowers ad attitudes and donation intentions. In a second study, they find that perceiving AI as more human-like weakens the negative effect, suggesting that the credibility penalty may be partly a function of the categorical distinction consumers draw between human and algorithmic communicators. Wortel, Vanwesenbeeck, and Tomas (2024) examined AI disclosures in Instagram advertising in a three-condition between-subjects experiment (N = 161). They found that AI disclosure reduced advertising attitude, while effects on brand attitude depended on AI aversion, and no significant effect was found for source credibility. These findings suggest that the consequences of AI disclosure are context-dependent, although the study did not examine emotional email communication or the present perceived authenticity measure.

Together, these studies suggest that AI disclosure can reduce perceived authenticity and trust in some communication contexts, and that rhetorical reframing does not always eliminate this response. What remains untested in the existing literature is whether providing substantive informational content about how AI was used for a specific individual, rather than merely re-wording the same disclosure, could produce a different outcome. This gap motivates the present study.

## **2.4 Personalization in Marketing**

Personalization has become one of the dominant pillars of modern marketing communications, with a growing body of research examining its antecedents, mechanisms, and boundary conditions (Chandra et al., 2022). Chandra, Verma, Lim, Kumar, and Donthu (2022), in a comprehensive bibliometric review of 383 publications,

define personalization in marketing as designing and producing communications in ways that resonate with individual customer preferences and behavioral histories. Personalized content is proposed to signal to the recipient that the sender has invested knowledge of them as a specific individual rather than treating them as a generic audience member. The review identifies relevance, reduced cognitive load, and consumer orientation as primary mechanisms through which personalization may increase engagement.

However, the relationship between personalization and consumer response is not uniformly positive. Aguirre, Mahr, Grewal, de Ruyter, and Wetzels (2015) identify what they call the personalization paradox: the same personalization cues that signal relevance to some consumers may signal surveillance and data exploitation to others. Their experiments suggest that the direction of personalization's effect depends critically on how the consumer interprets the data use underlying the communication. When personalization is perceived as delivering genuine relevance, it may increase engagement and trust. When it makes consumers aware that behavioral data were collected and used without a clear benefit, personalization may backfire by increasing privacy concerns and perceived intrusiveness.

This tension is directly relevant to the design of the present study. The personalized email content used here signals behavioral knowledge of the recipient. Consumers may interpret this signal as evidence of individual attention, or instead as evidence that behavioral information was used in an intrusive way. Aguirre et al. (2015) suggest that personalization may activate two competing consumer inferences: that the brand invested behavioral data as a signal of consumer orientation, or that the brand used personal data to produce an automated emotional message without genuine relational investment. Which inference dominates is theoretically underdetermined and may vary across consumers, which motivates treating personalization as a moderator to be empirically tested rather than assuming a uniform directional effect.

A final theoretical lens relevant to the interaction hypothesis comes from the algorithm appreciation literature. Logg, Minson, and Moore (2019) demonstrate that consumer responses to algorithmic involvement depend on the type of task: for tasks perceived as objective and data-driven, people tend to accept algorithmic judgment; for tasks requiring empathy or emotional understanding, they prefer human involvement. This finding suggests that the way a consumer categorizes a communication task may moderate whether algorithmic authorship is penalized. Specifically, framing the AI's role in data-driven terms might partially shift how consumers categorize the task, with potential implications for authenticity perceptions. These mechanisms are drawn on in Section 3 to develop directional prediction for H2.

### 3. Theoretical Framework and Hypotheses

The present study builds on Kirk and Givi's (2025) work by testing whether their reported AI-authorship pattern generalizes to a new sample and email context under generic content. It also tests content personalization as a moderator using a factorial design that differs from their design. Whereas Schilke and Reimann (2025) show that rhetorical reframing of the disclosure does not eliminate the negative response, the present study asks whether changing the informational content of the email itself can produce a different outcome.

#### The AI-Authorship Effect in Generic Email Conditions

When a consumer reads an emotional message and learns it was written by AI, the integrity of the communication becomes doubtful. Morhart et al. (2015) define integrity as the perception that a brand acts from genuine values rather than commercial motives. In the present context, consumers may infer that an AI system does not itself experience the gratitude or warmth expressed in the message, which can make the communication seem less authentic (Kirk & Givi, 2025).

Positive word of mouth and loyalty may depend partly on whether consumers infer genuine relational investment from the brand. If an emotional message appears less genuine, consumers may have less reason to respond with favorable behavioral intentions (Morhart et al., 2015; Kirk & Givi, 2025).

Kirk and Givi (2025) report this pattern across seven experiments, with perceived authenticity identified as an important explanatory construct. The present study tests the generic Human–Generic versus AI–Generic comparison in a loyalty email context. The comparison is intended as a replication-oriented test, not as a presumption that the prior effect must occur in the present sample.

**H1:** In generic email conditions, human-authored messages produce higher positive word of mouth, customer loyalty, and perceived authenticity than AI-authored messages.

## **Content Personalization as a Moderator of the AI-Authorship Effect**

For AI-authored messages, the authenticity problem is partly about inference. A consumer who knows AI wrote the message may assume it is a generic output and nothing about them specifically shaped it. A personalized email can change that assumption. When the message refers to pieces the recipient keeps returning to or collections they continue to explore, it may signal that the communication was adapted to the individual recipient. Logg et al. (2019) show that consumer responses to algorithmic involvement depend partly on the type of task. Consumers tend to be more accepting of algorithms in objective and data-processing tasks than in tasks involving empathy or emotional understanding. Personalization may therefore shift the perceived role of AI from producing generic emotional content toward processing information relevant to the consumer. The authenticity gap does not close entirely, but its basis changes. The message reads less like a template and more like an act of individual attention, addressing Morhart et al.'s (2015) integrity dimension more directly.

This argument draws on the consumer-orientation mechanism discussed by Chandra et al. (2022) and Aguirre et al. (2015). Personalization can signal that the sender has treated the recipient as an individual rather than as a member of a generic audience. In the AI context, this signal may partly shift the perceived role of the system from producing generic emotional content toward processing information relevant to the consumer. Because the present stimulus used mild and generalized behavioral references, the consumer-orientation interpretation was expected to be more likely than an intrusive surveillance interpretation. However, the personalization paradox means that either interpretation remains possible. This prediction is directional but treated as an open empirical question: the personalization paradox (Aguirre et al., 2015) leaves open the possibility that the same cues may instead amplify perceived inauthenticity if consumers interpret the behavioral references as automated surveillance without genuine relational investment.

For human-authored messages, this mechanism may have less room to operate. Consumers who believe that a person wrote the message already have a basis for

attributing genuine emotional experience to the sender. Personalization may therefore provide less additional benefit when the initial authenticity inference is already relatively favorable. This reasoning implies that personalization would be expected to produce a larger improvement in responses to AI-authored messages than in responses to human-authored messages, because AI-authored messages are presumed to begin from a lower baseline.

Two predictions follow. Within AI conditions, personalized messages should produce higher PWOM and loyalty than generic messages. Perceived authenticity is examined as a related secondary outcome. Across authorship conditions, the human–AI difference should be smaller for personalized emails than for generic emails, although it may not disappear completely.

**H2:** Personalization moderates the negative effect of AI authorship on positive word of mouth and loyalty, such that the difference between human- and AI-authored messages is smaller for personalized emails than for generic emails.

The predicted interaction implies that the AI–Personalized condition may produce more favorable responses than the AI–Generic condition. This comparison is a simple effect used to interpret the interaction; it does not independently establish moderation. Within human-authored conditions, the corresponding improvement is expected to be smaller or absent.

## 4. Experiment

### 4.1 Design

The study uses a  $2$  (authorship: human vs. AI)  $\times$   $2$  (email content: generic vs. personalized) between-subjects factorial design. Participants were randomly assigned to one of four conditions: Human–Generic, Human–Personalized, AI–Generic, or AI–Personalized. Across all four conditions, participants read the same scenario framing and the same base email format. Conditions differed in two respects: the content of the email body (generic vs. personalized) and the authorship signal in the signature (Sarah Mercer, Customer Relations Director vs. Lumiere AI Assistant). All other planned aspects of the stimulus were held constant. Together with random assignments, this design supported causal comparison of the assigned conditions, subject to the salience of the manipulations and the limitations of the measures.

A between-subjects structure was chosen to avoid carryover effects. Exposure to one condition would likely lead participants to interpret subsequent conditions as deliberate contrasts, increasing their sensitivity to the manipulation and introducing demand characteristics (Orne, 1962). In a between-subjects design, each participant responds to a single version of the email, producing a more realistic measure of a real-world consumer reaction.

### 4.2 Participants and Sampling

#### *Target Sample Size*

The original design targeted 160 participants, corresponding to approximately 40 participants per experimental condition. The target was determined using G\*Power 3.1 for a  $2 \times 2$  between-subjects ANOVA with  $\alpha = .05$  and planned power of .80. For H1, the planning benchmark was based on the perceived authenticity effect reported by Kirk and Givi (2025) in Study 5,  $\eta^2 = .18$ , corresponding to Cohen's  $f = .47$  and yielding more than

99% power at  $N = 160$ . For H2, no prior effect-size estimate was available for the authorship  $\times$  personalization interaction, so a conventional medium effect of  $f = .25$  was used as a planning benchmark. This required a minimum of 128 participants and yielded approximately 87% power at  $N = 160$ . The target of 160 also provided a buffer against attrition and incomplete responses.

The achieved sample consisted of 79 completed responses: Human–Generic ( $n = 14$ ), Human–Personalized ( $n = 23$ ), AI–Generic ( $n = 19$ ), and AI–Personalized ( $n = 23$ ). The smaller-than-planned sample resulted in wider confidence intervals and less precise estimates. A supplementary sensitivity analysis indicated that the final sample could detect approximately an H1 effect of  $d = 1.02$  and an H2 interaction of  $f = .32$  with approximately 80% power. These values describe the minimum effects detectable with the achieved sample, while confidence intervals around the observed effects are reported in Section 5.

The H1 value served as a literature-based benchmark for the expected authorship effect, while the final H1 analysis used direct Welch comparisons between the Human–Generic and AI–Generic conditions.

### ***Recruitment***

Participants were recruited through the researcher’s personal and professional networks via WhatsApp, LinkedIn, and social media channels, forming a convenience sample. Eligibility criteria are: (1) age 18 or older, and (2) sufficient English proficiency to read and understand the email stimulus. All 79 completed responses were included in the primary analytic sample. Authorship recall, personalization recall, and brand recall were treated as descriptive comprehension checks rather than automatic exclusion criteria. Three targeted supplementary analyses based on the relevant checks are reported in Appendix A.

### **4.3 Stimuli**

#### ***Fictional Brand***

The study uses a vignette methodology (Atzmuller & Steiner, 2010), in which participants read a written scenario designed to simulate a real-world marketing communication and respond as though it were genuine. A fictional fashion brand, Lumiere, was used across all conditions to prevent prior brand associations from affecting authenticity ratings. All participants read a scenario asking them to imagine they are a loyal customer of Lumiere and have just received an email from the company's customer relations department.

#### ***Email Stimuli***

Two versions of the email were created: a generic version and a personalized version. The two versions share the same structure, emotional tone, and level of warmth. They differed in three elements: the change from "someone chooses" to "you choose," the addition of mild and generalized behavioral references, and the closing phrase. The personalized stimuli version was developed with the assistance of a generative AI tool (ChatGPT, GPT-4o, OpenAI, 2024), consistent with the study's focus on AI-generated content. The researcher evaluated all drafts against the following criteria: emotional tone and warmth equivalent to the generic version; behavioral references that were plausible and generalizable across consumers without presupposing specific purchase details; and no factual claims inconsistent with the scenario. The final stimuli retained the same collective brand voice across authorship conditions. All revisions were made by the researcher, who reviewed and finalized the stimuli before inclusion.

**Generic email:**

*Subject: A heartfelt thank you*

*Dear Valued Customer,*

*We wanted to take a moment to say something simple, but something we mean deeply: thank you.*

*Every time someone chooses to shop with us, it reminds our entire team why we do what we do. Your continued trust means more to us than you may realize, and it inspires us to keep creating with care and intention.*

*We don't take your loyalty for granted, and we never will.*

*From all of us, thank you for being part of our story.*

*Warmly,*

*[Signature]*

**Personalized email:**

*Subject: A heartfelt thank you*

*Dear Valued Customer,*

*We wanted to take a moment to say something simple, but something we mean deeply: thank you.*

*Every time you choose to shop with us, from the pieces you keep returning to, to the collections you continue to explore with us, it reminds our entire*

*team why we do what we do. Your continued trust, built with us over time, means more to us than you may realize, and it inspires us to keep creating with care and intention.*

*We don't take your loyalty for granted, and we never will.*

*From all of us, thank you for continuing to be part of our story.*

*Warmly,*

*[Signature]*

The personalized version differs from the generic in three places: (1) "someone chooses" is replaced with "you choose"; (2) a behavioral reference is added ("from the pieces you keep returning to, to the collections you continue to explore with us"); and (3) the closing phrase "thank you for continuing to be part of our story" replaces "thank you for being part of our story." These changes were intended to signal individual knowledge of the recipient while keeping the emotional tone and warmth of the message constant.

### ***Experimental Conditions***

The four conditions are defined by the combination of email body and signature:

| <b>Condition</b>   | <b>Email Body</b>       | <b>Signature</b>                                                  |
|--------------------|-------------------------|-------------------------------------------------------------------|
| Human–Generic      | Generic email body      | Warmly, Sarah Mercer<br>Customer Relations<br>Director, Lumiere   |
| Human–Personalized | Personalized email body | Warmly, Sarah Mercer<br>Customer Relations<br>Director, Lumiere   |
| AI–Generic         | Generic email body      | Warmly, Lumiere AI<br>Assistant<br>Customer Relations,<br>Lumiere |
| AI–Personalized    | Personalized email body | Warmly, Lumiere AI<br>Assistant<br>Customer Relations,<br>Lumiere |

The human conditions carried no additional authorship disclosure beyond the human signature. The AI conditions signaled authorship through the signature “Lumiere AI Assistant.” This was a relatively subtle authorship cue, selected to represent a plausible customer communication format. Its limited salience is treated as an important limitation because only a minority of participants correctly recalled authorship.

The generic email stimuli in the present study and Kirk and Givi’s (2025) Study 5 both use emotional gratitude communication directed at loyal customers. However, the messages are not identical. Kirk and Givi’s Study 5 uses a more personal gratitude letter from a Sales Director at Avery Organics, whereas the present study uses a shorter loyalty email from the fictional fashion brand Lumiere. The messages therefore share a broad emotional and relational purpose but differ in wording, sender identity, brand context, and disclosure format.

The studies differ in how authorship was communicated. In Kirk and Givi’s human condition, the email contained no additional authorship disclosure. In their AI condition, the statement “This email was written by AI. You can learn more about our use of AI here.” was appended after the signature. In the present study, human authorship was signaled through the signature “Sarah Mercer, Customer Relations Director,” whereas AI authorship was signaled through “Lumiere AI Assistant.” The present study also varied the email body by adding a personalized version, whereas the message body was held constant across the human and AI conditions in Kirk and Givi’s Study 5. The present email and Kirk and Givi’s Study 5 both examine emotional gratitude communications directed toward existing customers, but they do not use identical messages. Kirk and Givi’s Study 5 used a more personal gratitude communication from a Sales Director at Avery Organics, whereas the present study used a loyalty email from the fictional fashion brand Lumiere. The messages therefore differed in wording, sender identity, brand context, and disclosure format. The present stimulus nevertheless retained collective and

human-like expressions such as “our entire team” and “it inspires us.” It therefore did not completely eliminate the possible tension between emotional language and an AI-attributed sender. The present study was designed as a contextual replication and extension rather than as a direct replication using identical stimuli. It retained the emotional gratitude context and the human-versus-AI authorship attribution contrast to examine whether a comparable pattern would emerge with a different brand, sender identity, disclosure format, sample, and email structure. The addition of content personalization further extended the design beyond Kirk and Givi’s study. Accordingly, the comparison with Kirk and Givi provides a contextual benchmark rather than a test of exact replication. H1 should therefore be interpreted as a test of whether a comparable human–AI difference emerged in the present context, not as an exact replication of Kirk and Givi’s stimulus and procedure.

#### **4.4 Procedure**

The survey was administered via Qualtrics, with built-in random assignment to the four experimental conditions. After providing informed consent, participants were directed to a scenario framing that asked them to imagine they are a loyal customer of Lumiere who has just received an email from the brand's customer relations department; they were asked to respond as though the situation was real. Participants then read the email stimulus in its assigned condition-specific form, with the subject line, body, and signature visible.

Following exposure to the stimulus, participants completed the primary outcome measures in the following order: positive word of mouth (three items), customer loyalty (two items), and perceived authenticity (four items). This sequencing captures behavioral intentions as the most immediate response to the email, with perceived authenticity included as a secondary outcome to examine whether the experimental conditions differed in their authenticity-related evaluations. Next, participants completed the single-

item AI familiarity measure, placed after outcome measures to prevent priming. Three manipulation check items were then presented in randomized order: authorship recall, personalization recall, and brand attention check. The survey closed with three demographic items: age, gender, and prior experience receiving AI-generated marketing emails. The survey was designed to take approximately five to six minutes to complete.

#### **4.5 Measures**

##### ***Primary Outcomes: Positive Word of Mouth and Customer Loyalty***

Positive word of mouth (PWOM) was measured using three items adapted from Alexandrov, Lilly, and Babakus (2013), consistent with Kirk and Givi (2025): "After receiving this email, I would say positive things about Lumiere," "After receiving this email, I would recommend Lumiere to others," and "After receiving this email, I would recommend Lumiere to someone else who seeks my advice." Items were rated on a 7-point scale (1 = strongly disagree, 7 = strongly agree) and averaged into a single composite ( $\alpha = .97-.98$  in Kirk & Givi, 2025).

Customer loyalty was measured using two items from Kirk and Givi (2025): "After receiving this email, I intend to remain a customer of Lumiere" (straight scored) and "After receiving this email, I would switch to a different brand for my fashion purchases" (reverse scored). Items were rated on a 7-point scale and averaged after reverse scoring. The inter-item correlation in the present sample was  $r = .222$  and is reported in the Results section.

##### ***Secondary Outcome: Perceived Authenticity***

Perceived authenticity was measured using the four-item bipolar scale validated by Napoli, Dickinson, Beverland, and Farrelly (2014) and used by Kirk and Givi (2025). Participants rated the brand's communication on four 7-point bipolar items: Inauthentic/Authentic, Not true to their own values/True to their own values, Lacking in

principles/Highly principled, and Compromising their values/Not compromising their values. Items were averaged into a single composite. Cronbach's alpha was .827 in the present sample. Kirk and Givi (2025) reported  $\alpha = .93$  for the corresponding scale in Study 5. Higher scores indicate greater perceived authenticity.

Although the authenticity scale includes items about being true to values and not compromising values, the brand's specific values are not stated in the email stimulus. This is the same design choice made by Kirk and Givi (2025) in their Study 5, which also used this scale with a generic brand email. Participants are therefore interpreting these items as general assessments of the communication's integrity rather than evaluations of specific known values. The scale is retained to enable direct comparison with Kirk and Givi's findings, and this limitation is acknowledged in Section 6.

### ***AI Familiarity***

A single item assessed participant familiarity with AI: "How familiar are you with Artificial Intelligence and its applications (such as ChatGPT)?" rated on a 5-point scale (1 = not at all familiar, 5 = very familiar), adapted from Schilke and Reimann (2025). This item was used descriptively to characterize the sample. It was not included as a covariate in the primary hypothesis tests.

### ***Manipulation Checks***

Three manipulation check items were included after all outcome measures, presented in randomized order: (1) authorship recall ("To the best of your recollection, who or what authored the email you just read?"; options: a human employee of the brand / artificial intelligence / I do not recall / the email did not specify); (2) personalization recall ("Based on what you read, was the email addressed and written specifically for you?"; options: yes / no / I do not recall); and (3) a brand attention check ("What was the name of the brand in the email?"; options: Lumiere / Aurora / Celeste / I do not recall).

## 5. Results

### 5.1 Sample and Manipulation Checks

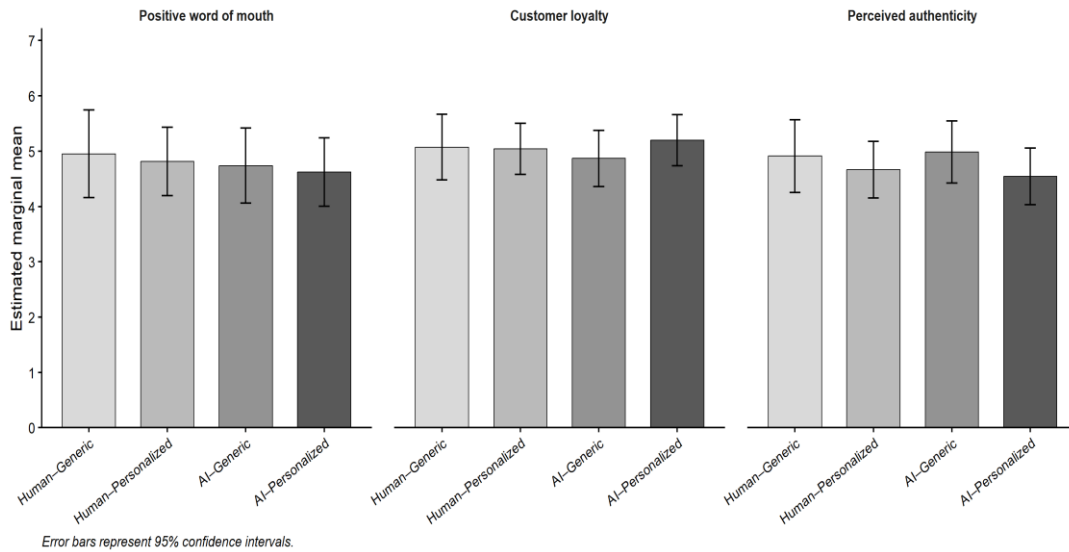
A total of 79 participants completed the survey and were included in the primary analysis. The final condition sizes were 14 participants in the Human–Generic condition, 23 in the Human–Personalized condition, 19 in the AI–Generic condition, and 23 in the AI–Personalized condition.

Positive word of mouth showed strong internal consistency, Cronbach's  $\alpha = .926$ . Perceived authenticity showed acceptable internal consistency,  $\alpha = .827$ . The two loyalty items were positive but weakly correlated,  $r = .222$ . The loyalty composite was retained because its construction was specified in advance, but the low inter-item correlation is treated as a measurement limitation.

The authorship manipulation was weakly recognized. Authorship was recalled correctly by 3 of 14 participants in Human–Generic (21.4%), 9 of 23 in Human–Personalized (39.1%), 5 of 19 in AI–Generic (26.3%), and 10 of 23 in AI–Personalized (43.5%), for 27 of 79 participants overall (34.2%).

For the personalization check, 28 of 33 participants assigned to generic emails correctly identified the message as generic (84.8%). In the personalized conditions, only 7 of 46 participants correctly identified the message as personalized (15.2%). Brand recall was substantially stronger, with 76 of 79 participants correctly identifying Lumiere as the brand (96.2%). These checks were not used as exclusion criteria for the primary analysis.

## 5.2 Descriptive Statistics



**Figure 1.** Estimated marginal means for positive word of mouth, customer loyalty, and perceived authenticity across the four experimental conditions. Error bars represent 95% confidence intervals.

For positive word of mouth, the observed mean was highest in the Human–Generic condition ( $M = 4.95$ ), followed by the Human–Personalized condition ( $M = 4.81$ ), the AI–Generic condition ( $M = 4.74$ ), and the AI–Personalized condition ( $M = 4.62$ ). Thus, the human-authored conditions had higher observed PWOM means than the corresponding AI-authored conditions under both generic and personalized content.

For customer loyalty, the Human–Generic condition had a mean of 5.07 and the Human–Personalized condition a mean of 5.04. The AI–Generic condition had a mean of 4.87, whereas the AI–Personalized condition had the highest observed loyalty mean at 5.20. Thus, the human-authored condition had the higher loyalty mean under generic content, while the AI-authored condition had the higher loyalty mean under personalized content.

For perceived authenticity, the AI–Generic condition had the highest observed mean at 4.99, followed by the Human–Generic condition at 4.91. Under personalized content, the Human–Personalized condition had a mean of 4.66 and the AI–Personalized condition a mean of 4.54. Thus, the AI-authored condition had the slightly higher authenticity mean under generic content, whereas the human-authored condition had the higher authenticity mean under personalized content.

Comparing generic and personalized content within each authorship condition, personalization was associated with lower observed PWOM under both human authorship, from 4.95 to 4.81, and AI authorship, from 4.74 to 4.62. Personalization was also associated with lower perceived authenticity under both human authorship, from 4.91 to 4.66, and AI authorship, from 4.99 to 4.54. Loyalty followed a different pattern: it changed only slightly under human authorship, from 5.07 to 5.04, but increased descriptively under AI authorship, from 4.87 to 5.20.

**Table 1**

Manipulation-Check Accuracy by Condition

| Condition          | n  | Authorship correct n (%) | Personalization correct n (%) | Brand correct n (%) |
|--------------------|----|--------------------------|-------------------------------|---------------------|
| Human–Generic      | 14 | 3 (21.4%)                | 13 (92.9%)                    | 14 (100%)           |
| Human–Personalized | 23 | 9 (39.1%)                | 2 (8.7%)                      | 21 (91.3%)          |
| AI–Generic         | 19 | 5 (26.3%)                | 15 (78.9%)                    | 18 (94.7%)          |
| AI–Personalized    | 23 | 10 (43.5%)               | 5 (21.7%)                     | 23 (100%)           |
| Total              | 79 | 27 (34.2%)               | 35 (44.3%)                    | 76 (96.2%)          |

### 5.3 AI Authorship in Generic Email Conditions

H1 predicted higher outcomes for Human–Generic than AI–Generic messages. The prespecified direct comparisons were conducted using two-sided Welch independent-samples t-tests, with Cohen’s d reported as the standardized mean difference.

For PWOM, the observed mean was higher in the Human–Generic condition ( $M = 4.95$ , 95% CI [4.27, 5.63]) than in the AI–Generic condition ( $M = 4.74$ , 95% CI [3.91, 5.56]). The mean difference was 0.22 points, 95% CI [-0.81, 1.24],  $t(30.90) = 0.43$ ,  $p = .671$ ,  $d = .14$ . For customer loyalty, the observed mean was also higher in the Human–Generic condition ( $M = 5.07$ , 95% CI [4.51, 5.64]) than in the AI–Generic condition ( $M = 4.87$ , 95% CI [4.31, 5.43]). The mean difference was 0.20 points, 95% CI [-0.56, 0.97],  $t(30.40) = 0.54$ ,  $p = .591$ ,  $d = .19$ . For perceived authenticity, the AI–Generic condition had a slightly higher observed mean ( $M = 4.99$ , 95% CI [4.34, 5.63]) than the

Human–Generic condition ( $M = 4.91$ , 95% CI [4.30, 5.52]). The mean difference was  $-0.08$  points, 95% CI [-0.92, 0.77],  $t(30.83) = -0.18$ ,  $p = .856$ ,  $d = -.06$ .

The observed pattern therefore favored human authorship for PWOM and loyalty, but not for perceived authenticity. However, the confidence intervals were wide and all three formal comparisons were nonsignificant. The results do not provide statistically reliable evidence for the predicted human-authorship advantage. H1 was therefore not supported.

#### **5.4 Personalization as a Moderator**

H2 predicted that personalization would reduce the difference between human- and AI-authored messages. The primary test was the authorship  $\times$  personalization interaction in the  $2 \times 2$  factorial ANOVA. The observed condition means suggested different patterns across the three outcomes.

For PWOM, the Human–Generic condition had a higher observed mean than the AI–Generic condition, 4.95 versus 4.74. Under personalized content, the Human–Personalized condition also had a higher observed mean than the AI–Personalized condition, 4.81 versus 4.62. The observed human–AI difference was therefore similar across generic and personalized content, approximately 0.21 and 0.19 points, respectively.

For customer loyalty, the Human–Generic condition had a higher observed mean than the AI–Generic condition, 5.07 versus 4.87. Under personalized content, however, the AI–Personalized condition had a higher observed mean than the Human–Personalized condition, 5.20 versus 5.04. The observed human–AI difference therefore changed direction between generic and personalized content.

For perceived authenticity, the AI–Generic condition had a slightly higher observed mean than the Human–Generic condition, 4.99 versus 4.91. Under personalized content, the Human–Personalized condition had a higher observed mean than the AI–Personalized

condition, 4.66 versus 4.54. The direction of the observed human–AI difference therefore also changed between generic and personalized content.

The factorial ANOVA did not provide statistically reliable evidence for these observed changes. For PWOM, there was no significant main effect of authorship,  $F(1, 75) = 0.35$ ,  $p = .556$ , partial  $\eta^2 = .005$ , or personalization,  $F(1, 75) = 0.14$ ,  $p = .711$ , partial  $\eta^2 = .002$ . The authorship  $\times$  personalization interaction was close to zero, estimate = 0.03, 95% CI  $[-1.33, 1.39]$ ,  $F(1, 75) = 0.002$ ,  $p = .968$ , partial  $\eta^2 < .001$ .

For customer loyalty, there was no significant main effect of authorship,  $F(1, 75) = 0.01$ ,  $p = .921$ , partial  $\eta^2 < .001$ , or personalization,  $F(1, 75) = 0.34$ ,  $p = .561$ , partial  $\eta^2 = .005$ . The authorship  $\times$  personalization interaction was positive but uncertain, estimate = 0.36, 95% CI  $[-0.67, 1.38]$ ,  $F(1, 75) = 0.48$ ,  $p = .490$ , partial  $\eta^2 = .006$ .

For perceived authenticity, there was no significant main effect of authorship,  $F(1, 75) = 0.006$ ,  $p = .939$ , partial  $\eta^2 < .001$ , or personalization,  $F(1, 75) = 1.49$ ,  $p = .225$ , partial  $\eta^2 = .020$ . The authorship  $\times$  personalization interaction was negative but uncertain, estimate =  $-0.20$ , 95% CI  $[-1.32, 0.93]$ ,  $F(1, 75) = 0.12$ ,  $p = .730$ , partial  $\eta^2 = .002$ .

The simple effect within the AI-authored conditions was examined only to interpret the interaction. Within the AI-authored conditions, personalization was associated with a change of  $-0.11$  points in PWOM, from 4.74 in the AI–Generic condition to 4.62 in the AI–Personalized condition, 95% CI  $[-1.03, 0.80]$ ,  $p = .806$ . For loyalty, personalization was associated with an increase of 0.33 points, from 4.87 to 5.20, 95% CI  $[-0.36, 1.02]$ ,  $p = .347$ . For perceived authenticity, personalization was associated with a decrease of 0.44 points, from 4.99 to 4.54, 95% CI  $[-1.20, 0.32]$ ,  $p = .249$ .

Overall, the observed pattern differed across outcomes. Personalization left the human–AI difference almost unchanged for PWOM, changed the direction of the observed loyalty difference, and also changed the direction of the observed authenticity difference.

However, the interaction confidence intervals included zero for all outcomes, and none of the interaction tests were statistically significant. The findings therefore show outcome-specific descriptive differences but do not provide statistically reliable evidence that personalization moderated the effect of authorship. H2 was not supported.

### **5.5 Summary of Findings**

Across the generic email conditions, the Human–Generic condition had higher observed PWOM and loyalty means than the AI–Generic condition, whereas the AI–Generic condition had a slightly higher observed perceived authenticity mean. Across the personalized email conditions, the Human–Personalized condition had higher observed PWOM and perceived authenticity means, whereas the AI–Personalized condition had the higher observed loyalty mean. Comparing generic and personalized content within each authorship condition, personalization was associated with lower PWOM under both authorship conditions, lower perceived authenticity under both authorship conditions, and higher loyalty only under AI authorship. The formal H1 and H2 tests did not establish that these observed differences were statistically reliable. The descriptive pattern nevertheless provides a basis for comparing the present study with Kirk and Givi’s (2025) findings in the General Discussion.

## 6. General Discussion

This study examined whether content personalization moderates consumer responses to AI-authored emotional marketing communications. The study produced clear descriptive differences across the four conditions, although the formal statistical tests did not establish that these differences were reliable population effects. Under generic content, the Human–Generic condition had higher observed PWOM and loyalty means than the AI–Generic condition, whereas the AI–Generic condition had a slightly higher observed perceived authenticity mean. Under personalized content, the Human–Personalized condition had higher observed PWOM and perceived authenticity means, whereas the AI–Personalized condition had the higher observed loyalty mean. Within the AI-authored conditions, personalization was associated with lower PWOM, higher loyalty, and lower perceived authenticity.

The direct Human–Generic versus AI–Generic comparisons were not statistically significant, and the authorship  $\times$  personalization interaction was not significant for any outcome. The results should therefore be interpreted as an informative but inconclusive test of when and how AI authorship and content personalization influence responses to emotional marketing communication. The observed differences were outcome-specific and did not form a consistent pattern supporting either a general AI-authorship penalty or a general benefit of personalization.

### 6.1 A Weaker or Context-Dependent AI-Authorship Effect

One possible interpretation is that the AI-authorship effect is weaker or more context-dependent than suggested by prior research. Kirk and Givi's (2025) Study 5 reported higher PWOM, loyalty, and perceived authenticity for human-authored than AI-authored emotional communication. Their reported mean differences were approximately 1.10 points for PWOM, 0.98 points for loyalty, and 1.39 points for perceived authenticity. In the present study's generic conditions, the corresponding observed differences were 0.22

points for PWOM, 0.20 points for loyalty, and  $-0.08$  points for perceived authenticity. The present study therefore did not reproduce the same substantive pattern, particularly for perceived authenticity, where the AI–Generic mean was slightly higher than the Human–Generic mean.

This comparison is theoretically informative, although the two studies should not be treated as identical replications. Kirk and Givi’s Study 5 used an emotional gratitude letter from a Sales Director at Avery Organics, whereas the present study used a shorter loyalty email from the fictional fashion brand Lumiere. Both studies examined emotional gratitude communication directed at loyal customers, but the messages differed in wording, sender identity, brand context, and authorship cue. The AI-authorship effect may therefore depend on the emotional intensity of the message, the perceived relationship between the sender and the recipient, the identity of the apparent author, or the way AI involvement is communicated.

The present findings do not demonstrate that the AI-authorship effect does not exist. The confidence intervals remain wide, and the achieved sample was smaller than planned. At the same time, the observed generic-condition differences were substantially smaller and less consistent than those reported by Kirk and Givi (2025), and the perceived-authenticity difference was slightly reversed in direction. The results are therefore consistent with the possibility that the effect is weaker or more context-dependent in the present setting, although the imprecise estimates and weak recognition of the manipulations prevent a definitive distinction between a genuinely weaker effect and insufficiently sensitive measurement.

## **6.2 Salience of the Authorship and Personalization Manipulations**

A second possible explanation concerns the salience of the manipulations. Authorship was correctly recalled by 27 of 79 participants, or 34.2%. This does not necessarily mean that participants simply failed the manipulation check. It may indicate that the authorship

cue was too subtle, that participants did not consider the signature information sufficiently important to retain, or that they interpreted the sender identity differently from the intended manipulation. If participants did not psychologically register the difference between “Sarah Mercer, Customer Relations Director” and “Lumiere AI Assistant,” the assigned authorship contrast would be less likely to produce a strong response difference.

Personalization was recognized even less consistently. Only 7 of the 46 participants assigned to personalized emails correctly identified the message as personalized. This may indicate that the behavioral references were not interpreted as individualized, that the personalization was too mild to attract attention, or that participants did not retain the relevant information when answering the check. Because the check was administered after exposure, the responses cannot distinguish whether participants failed to notice the cue or noticed it but did not retain it.

Weak recognition of personalization manipulation may have reduced the observed interaction. The assigned personalized and generic conditions may not have corresponded closely to participants perceived experiences. The nonsignificant H2 interaction may therefore reflect the limited salience of the present manipulation rather than demonstrating that this type of personalized content cannot moderate responses to AI-attributed communication. At the same time, the present data cannot determine which of these explanations is correct.

### **6.3 Differences from Previous Experimental Paradigms**

A third possible explanation concerns the authorship manipulation and disclosure format. In Kirk and Givi’s (2025) Study 5, the human condition contained no additional authorship disclosure, whereas the AI condition appended a statement indicating that the email had been written by AI. In the present study, human authorship was signaled through the signature “Sarah Mercer, Customer Relations Director,” whereas AI

authorship was signaled through “Lumiere AI Assistant.” The present study therefore used a different and potentially more direct authorship cue.

This difference may be theoretically relevant because Kirk and Givi’s broader findings suggest that consumer responses depend partly on who appears to sign AI-generated communication. In their Study 4, the AI-authorship effect was weaker when the AI signed the communication directly rather than generating the message while a human signed it. In the present study, the AI condition was also signed directly by “Lumiere AI Assistant.” It is therefore possible that the direct AI signature made the communication more coherent and reduced the perceived inconsistency between AI authorship and the identity of the apparent sender. This interpretation is consistent with Kirk and Givi’s findings, but the present study did not independently manipulate direct versus human signing and cannot test this explanation directly.

The present stimulus also does not eliminate the possible inconsistency between emotional language and AI authorship. The email retains collective and human-like expressions, including “we mean deeply,” “our entire team,” and “it inspires us.” The present design should therefore not be described as a completely clean test in which AI authorship is separated from human-like emotional language. Instead, it tests a different combination of message content, brand context, sender identity, and authorship cue.

The personalization manipulation introduced an additional difference from Kirk and Givi’s study. The present personalized version used mild and generalized behavioral references rather than explicit purchase-history information. These references may not have been sufficiently salient to create either a strong consumer-orientation inference or a strong privacy and intrusiveness concern. In other words, the manipulation may not have activated either side of the personalization paradox strongly enough to produce a measurable interaction. This remains a possibility because very few participants interpreted the message as personalized.

## **6.4 Practical Implications**

The practical implications of these results depend on which explanation best accounts for the observed pattern. It is possible that AI authorship has a weaker effect in this type of customer email than in some previous studies. It is also possible that many participants did not notice or remember the authorship information. Because the present study cannot distinguish between these explanations, the results do not show that companies can use AI in emotional customer communication without concern.

The results also do not show that personalization is generally ineffective. In this study, the personalization was mild and used general behavioral references. Participants may not have noticed these references or may not have seen them as personal. Personalization can make a message feel more relevant, but it can also make people worry about how their data were used or feel that the message is intrusive (Aguirre et al., 2015). Companies should therefore test whether customers notice the personalization and experience it as useful before assuming that it will improve responses to AI-authored messages.

The study also does not support a general recommendation either to avoid AI in all emotional communication or to use it without caution. A more appropriate recommendation is that companies should test different ways of communicating AI involvement and different types of personalization with their target customers. They should measure not only customer responses, but also whether people noticed the AI information, understood the personalization, and experienced the message as authentic, relevant, or intrusive.

## **6.5 Theoretical Contributions**

The study's theoretical contribution is to test whether content personalization can alter responses to an AI-authored emotional message. The observed results do not support a general conclusion that personalization improves responses to AI-authored communication. Instead, the descriptive pattern varied across outcomes: personalization was associated with lower PWOM and perceived authenticity within the AI-authored

conditions, but higher loyalty. The interaction tests did not establish that these changes were statistically reliable.

The study also illustrates the importance of distinguishing between the assigned manipulation and the manipulation as perceived by participants. A personalization manipulation can be present in email but fail to produce a clear perception of personalization. Similarly, an authorship cue can be visible in the signature without becoming psychologically salient to the recipient.

Finally, the comparison with Kirk and Givi (2025) suggests that the AI-authorship effect may depend on disclosure format, sender identity, emotional language, and the relationship between the apparent author and the message content. The present study does not identify which of these factors explains the different descriptive pattern, but it shows why they should be considered when generalizing the AI-authorship effect across marketing contexts.

## **6.6 Limitations**

Several limitations should be noted. First, the study used a hypothetical vignette and a single exposure to one fictional fashion brand. Vignette experiments provide experimental control, but responses to a written scenario may not represent behavior toward a real brand or an actual email (Atzmüller & Steiner, 2010).

Second, the manipulations were weakly recognized. Only 34.2% of participants correctly recalled authorship, and only 15.2% of participants in the assigned personalized conditions correctly recalled the personalization manipulation. This creates uncertainty about whether the intended causal contrasts were salient to participants. The targeted recall analyses do not resolve this concern because their retained samples were extremely small.

Third, the final sample was a convenience sample of 79 participants, with unequal cell sizes, compared with the original target of 160 participants. The smaller-than-planned sample resulted in wider confidence intervals and less precise estimates. A supplementary sensitivity analysis indicated that the achieved design could detect approximately an H1 effect of  $d = 1.02$  and an H2 interaction of  $f = .32$  with approximately 80% power. The nonsignificant results therefore cannot be interpreted as evidence of equivalence or as proof that the effects are absent.

Fourth, the AI authorship signal was subtle, appearing in the signature as “Lumiere AI Assistant,” whereas prior work has used more explicit disclosure formats (Kirk & Givi, 2025). This difference may limit direct comparison with prior effect sizes. The sample also had relatively high AI familiarity ( $M = 4.18$  on a 5-point scale), which may limit generalizability to less familiar audiences.

Fifth, loyalty was measured with two items whose inter-item correlation was low in the present sample,  $r = .222$ . The prespecified composite was retained, but the loyalty findings should be interpreted cautiously.

Sixth, all dependent variables were self-reported intentions and judgments rather than observed behavior. Reported loyalty, willingness to recommend, and perceived authenticity may not translate directly into actual purchasing, switching, or word-of-mouth behavior. Future research should therefore examine behavioral indicators such as click-through behavior, repeat purchase, brand switching, or actual recommendations.

Finally, the study used one emotional loyalty email and one mild, generalized form of personalization. Stronger, more specific, or more salient personalization could produce different relevance, authenticity, privacy, or intrusiveness responses.

## **6.7 Future Research Directions**

Future research should use larger, preregistered samples with sufficient power for both authorship contrasts and interactions, while recruiting participants with varied levels of AI familiarity, privacy concern, and digital engagement. Such designs would improve precision and allow theoretically motivated heterogeneity tests.

Stronger and more systematically varied personalization manipulations should be tested, including name-based personalization, product-specific references, and explicit purchase-history acknowledgment. Manipulation checks should assess both whether personalization was noticed and whether it was perceived as relevant, helpful, intrusive, or surveillance-like.

Future studies should compare subtle and explicit AI disclosure formats and examine whether disclosure format interacts with personalization. This would distinguish the effects of authorship attribution from the effects of how prominently the attribution is communicated (Kirk & Givi, 2025; Schilke & Reimann, 2025).

Privacy concern and perceived intrusiveness should be measured explicitly as potential moderators or mechanisms, as predicted by the personalization paradox (Aguirre et al., 2015). Researchers should also distinguish generalized behavioral references, such as those used here, from highly specific data disclosures.

Finally, longitudinal and cross-cultural studies should examine repeated exposure, habituation to AI-authored communication, actual purchasing or switching, and differences across markets and consumer segments. These designs would clarify whether responses observed after one hypothetical email persist in ongoing brand relationships and whether they translate into observable consumer behavior.

## **Data and Analysis Materials**

An anonymized dataset, data dictionary, final analysis scripts, statistical outputs, figure files, survey documentation, and analysis notes are available in an OSF project: <https://osf.io/tbquh/overview>

## 7. Conclusion

This study tested whether content personalization changes consumer responses to AI-authored emotional marketing communication. The results showed different patterns across the three outcomes and experimental conditions. Under generic content, the Human–Generic condition had higher observed PWOM and loyalty means than the AI–Generic condition, while the AI–Generic condition had a slightly higher perceived authenticity mean. Under personalized content, the Human–Personalized condition had higher observed PWOM and perceived authenticity means, whereas the AI–Personalized condition had the higher observed loyalty mean. Personalization therefore did not produce a consistent directional pattern across outcomes. Compared with Kirk and Givi’s (2025) Study 5, the observed generic-condition differences were also substantially smaller and less consistent, particularly for perceived authenticity, where the direction was slightly reversed.

The formal H1 comparisons and H2 interaction tests were not statistically significant, and the confidence intervals included values compatible with both positive and negative effects. The findings should therefore not be interpreted as evidence that AI authorship or personalization has no effect. Rather, they indicate that the predicted AI-authorship penalty and its proposed moderation through personalization did not generalize clearly to the present sample, stimulus, authorship cue, and communication context. The comparison with Kirk and Givi (2025) suggests that the magnitude of the AI-authorship effect may depend on disclosure format, sender identity, emotional language, manipulation salience, and the perceived relationship between the author and the message. The study provides a transparent test of these possibilities and identifies stronger manipulations, larger samples, behavioral outcomes, and explicit measures of privacy and intrusiveness as priorities for future research.

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## **Appendix A. Targeted Supplementary Analyses**

The primary analyses retained all 79 completed responses and did not automatically exclude participants based on the post-stimulus comprehension checks. Three targeted supplementary analyses were conducted to examine whether the estimates changed among participants who correctly recalled the manipulation relevant to each comparison. The H1 comparison was restricted to participants with correct authorship recall, the AI simple effect was restricted to participants with correct personalization recall, and the H2 interaction was restricted to participants with correct recall of both authorship and personalization. Because the retained samples were very small, these analyses are highly imprecise and are not confirmatory. The full sample analyses remain the primary analyses.

### **H1: Correct authorship recall**

Among the eight participants who correctly recalled authorship in the generic conditions (Human–Generic  $n = 3$ ; AI–Generic  $n = 5$ ), the Human–Generic minus AI–Generic differences were 0.58 for PWOM ( $p = .595$ ,  $d = .336$ ), 0.73 for loyalty ( $p = .262$ ,  $d = .914$ ), and  $-0.10$  for perceived authenticity ( $p = .891$ ,  $d = -.084$ ). These estimates are highly uncertain because the retained groups were very small.

### **AI simple effect: Correct personalization recall**

Among the 20 participants who correctly recalled personalization in the AI conditions (AI–Personalized  $n = 5$ ; AI–Generic  $n = 15$ ), the AI–Personalized minus AI–Generic differences were  $-1.36$  for PWOM ( $p = .149$ ,  $d = -.924$ ),  $-0.67$  for loyalty ( $p = .149$ ,  $d = -.596$ ), and  $-0.38$  for perceived authenticity ( $p = .647$ ,  $d = -.285$ ). These results are supplementary and do not independently establish moderation.

### **H2 interaction: Correct authorship and personalization recall**

Among the 12 participants who correctly recalled both authorship and personalization, the authorship  $\times$  personalization interactions were not significant for PWOM,  $F(1, 8) = 0.40$ ,  $p = .546$ ; loyalty,  $F(1, 8) = 0.005$ ,  $p = .943$ ; or perceived authenticity,  $F(1, 8) = 1.33$ ,  $p = .283$ . These analyses have very limited precision because the retained sample is small

and the cells are sparse. They should not be interpreted as confirming or disconfirming the primary H2 result.

Overall, these targeted analyses illustrate how the estimates behaved under stricter manipulation-recognition conditions, but their small and uneven samples prevent strong inferential conclusions. The full sample analyses remain the basis for the primary hypothesis conclusions.

## **Appendix B. Analysis Materials and Reproducibility**

The anonymized data, data dictionary, survey materials, analysis code, statistical outputs, figure files, and additional documentation associated with this study are available through the project's Open Science Framework repository: <https://osf.io/tbquh/overview>

The repository contains the materials required to reproduce the data preparation, descriptive analyses, primary hypothesis tests, targeted supplementary analyses, sensitivity analysis, and figure generation. Only anonymized and appropriate-to-share materials are included. The raw Qualtrics export and files containing direct identifiers or unnecessary metadata are not included in the public repository. The repository also includes `Claude_AI_usage_disclosure.txt` and `Dust_AI_usage_disclosure.txt` under the documentation folder, a verbatim transcript of the analysis-related GenAI exchanges, including the drafting and troubleshooting of the R code.

The analysis package contains the following seven R scripts, which should be executed in sequence:

1. `01_import_clean_score.R` imports the Qualtrics data, derives the experimental conditions from the authorship and personalization manipulation variables, checks the exported condition information, reverse scores the loyalty switching item, calculates the composite measures, and saves the processed analysis dataset.
2. `02_descriptives_reliability.R` calculates reliability estimates, sample descriptives, condition-level means and confidence intervals, condition sizes, and manipulation-check summaries.

3. 03\_primary\_anova\_contrasts.R estimates the  $2 \times 2$  factorial ANOVA models, produces estimated marginal means, and calculates the planned contrasts. The authorship  $\times$  personalization interaction is the primary test of the moderation hypothesis.
4. 04\_h1\_independent\_ttests.R conducts the direct two-sided Welch independent-samples comparisons between the Human–Generic and AI–Generic conditions for positive word of mouth, customer loyalty, and perceived authenticity. Cohen’s  $d$  is reported as the standardized mean difference.
5. 05\_targeted\_robustness\_checks.R conducts the supplementary analyses based on correct authorship recall, correct personalization recall, and correct recall of both manipulations. These analyses are supplementary because the retained samples are very small.
6. 06\_power\_analysis.R calculates sensitivity thresholds using the final analyzed sample. It reports the approximate H1 and H2 effect sizes detectable with 80% power.
7. 07\_figure1\_condition\_means.R generates the estimated marginal means figure with 95% confidence intervals for positive word of mouth, customer loyalty, and perceived authenticity.

The primary analysis uses all 79 completed responses. The loyalty switching item is reverse scored as 8 minus the original response before the loyalty composite is calculated. Conditions are derived from the authorship and personalization manipulation variables rather than manually trusted condition labels. The final anonymized dataset, R scripts, statistical outputs, figure files, survey documentation, and analysis notes are available in the OSF project.